

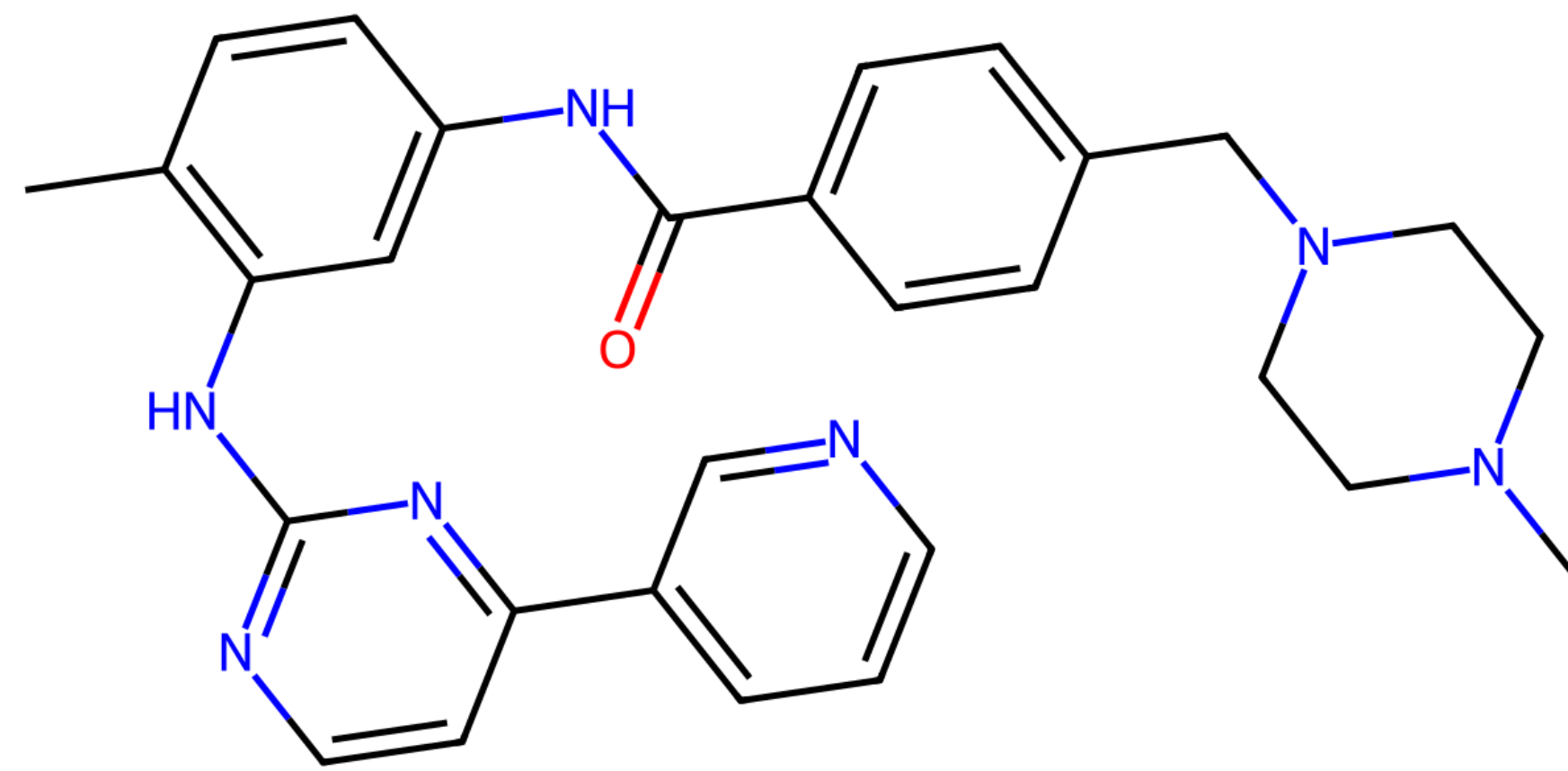
ChemSim-GPU

GPU-accelerated Tanimoto similarity for drug-discovery virtual screening

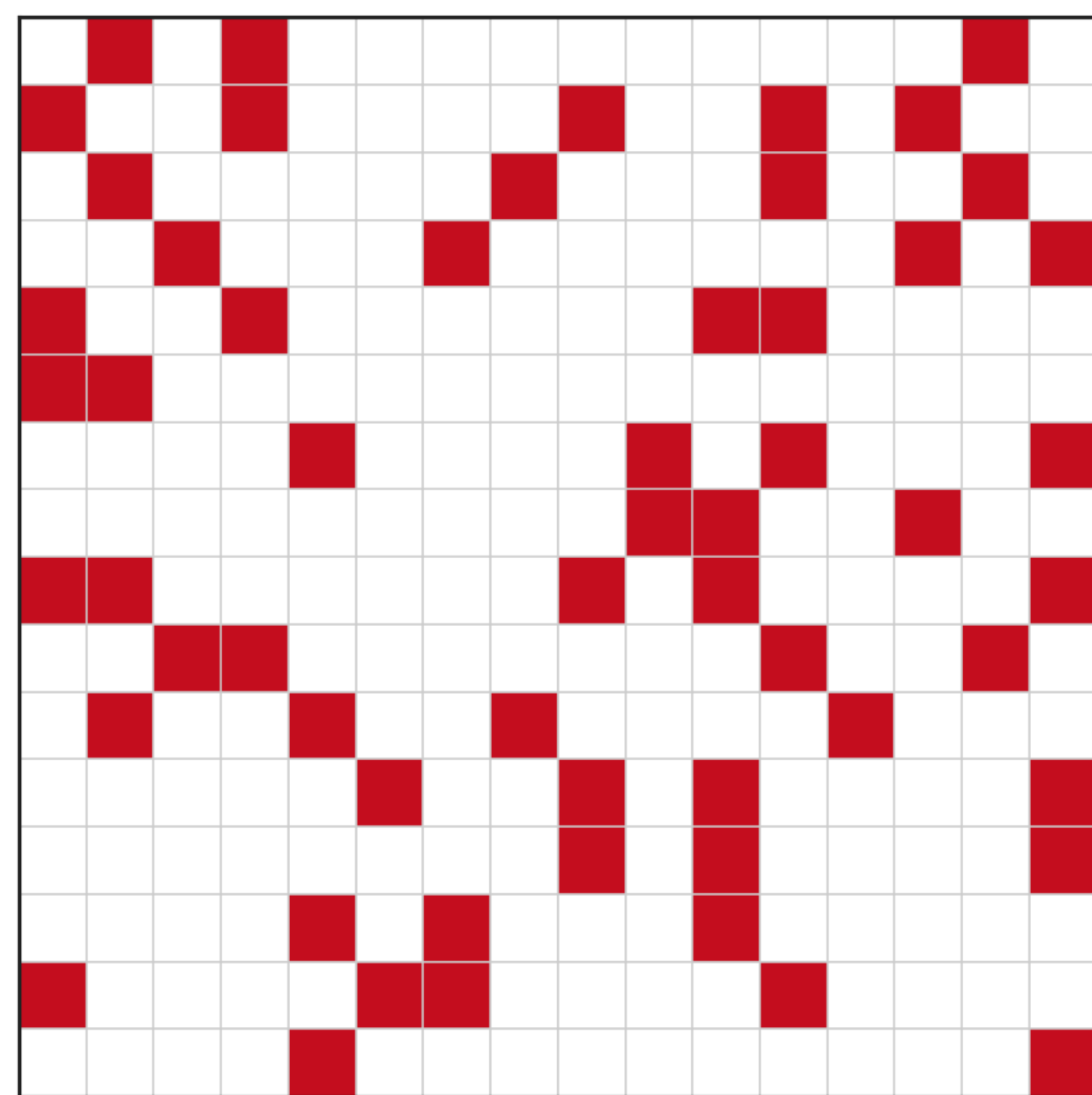
Kiarash Mokhtari Dizaji GPU Computing · TU Berlin · Summer Semester 2026

PROBLEM & MOTIVATION

Virtual screening looks through a large compound library for molecules similar to a known active one. Each molecule is turned into a binary fingerprint whose bits mark structural features.



Imatinib, drawn with RDKit



Its Morgan fingerprint (shown at 256 bits; benchmarks use 1024/2048)

Similarity is the Tanimoto coefficient:

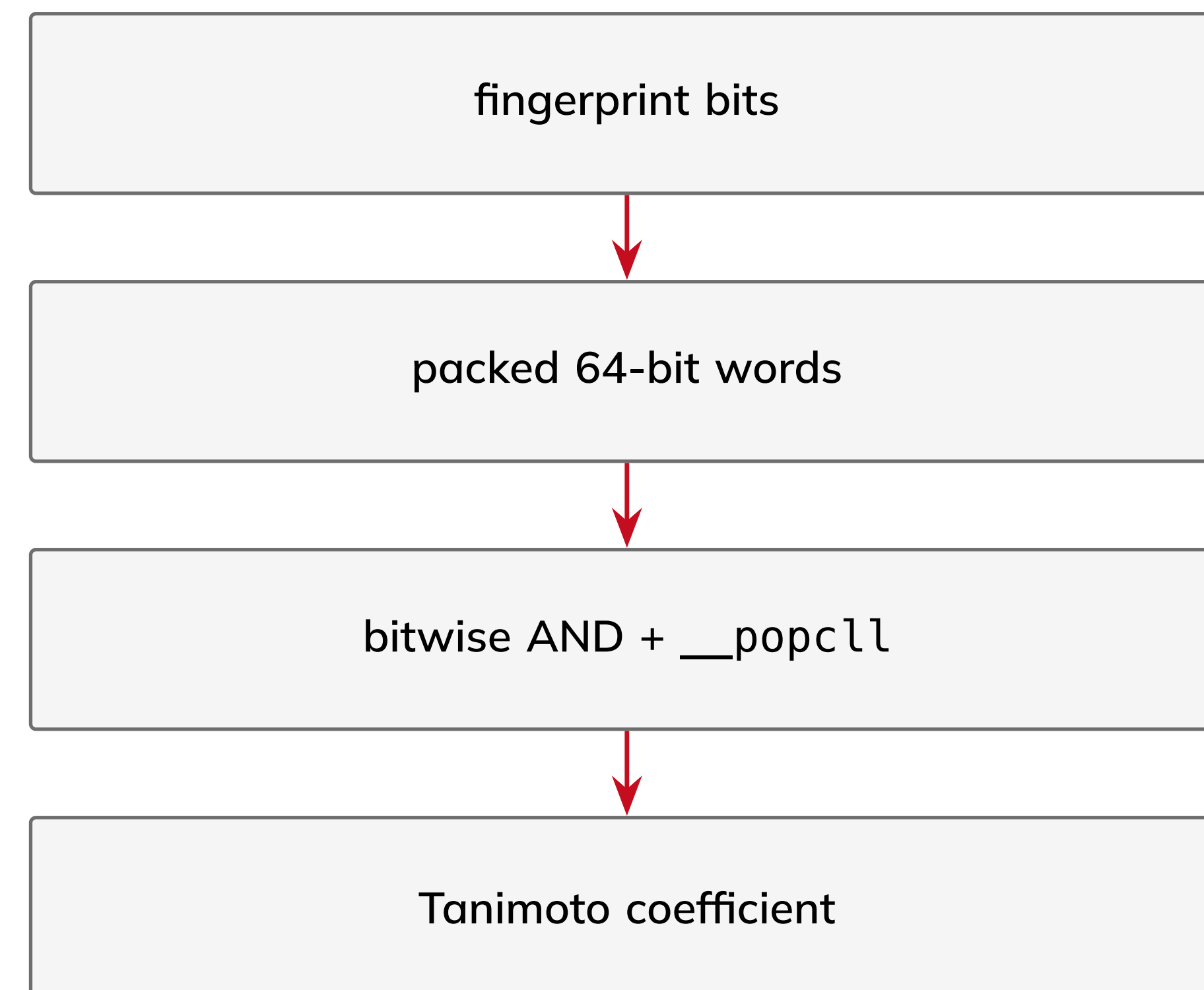
$$T = \frac{c}{a + b - c}$$

where a, b, c are the popcounts of A, B, and A AND B.

WHY IT FITS A GPU

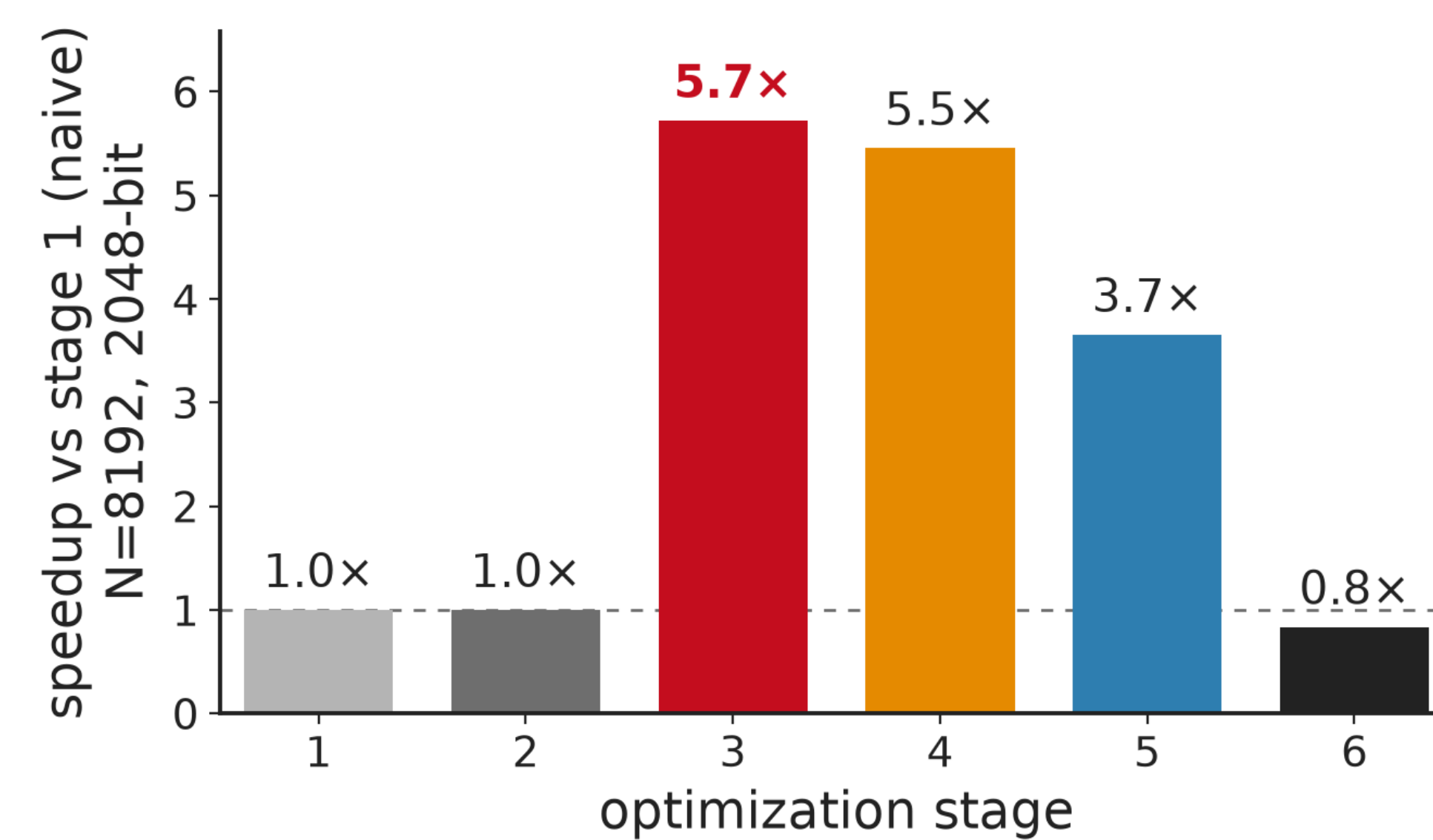
- $O(N^2)$ output cells, all independent.
- No thread divergence: every thread runs the same AND plus popcount.
- Hardware `__popc` counts set bits in one instruction.
- Like a binary GEMM with popcount instead of multiply-add.

METHOD



OPTIMIZATION LADDER

Each stage is a separate kernel, measured against the naive baseline (N=8192, 2048-bit).



#	Technique	What changed
1	Naive baseline	one thread per cell, software popcount
2	Bit-pack + __popc	hardware popcount; still memory-bound
3	Coalescing (SoA)	adjacent threads read adjacent words
4	Shared-mem tiling	reuse fingerprints on-chip
5	Thread coarsening	reuse query word across columns
6	Warp-shuffle	lanes split words, __shfl

Coalescing is the big win. Coarsening and warp-shuffle regress at this size, the honest result that shows the memory-bound ceiling.

CORRECTNESS & VALIDATION

- Every stage checked element-wise against a CPU reference over 4,000,000 pairs (2000×2000); max abs error 0.0.
- RDKit BulkTanimotoSimilarity is the reference standard.
- Edge cases: T(0,0)=0, identical=1, disjoint=0.
- Counts are exact integers, so GPU and CPU agree to float rounding.

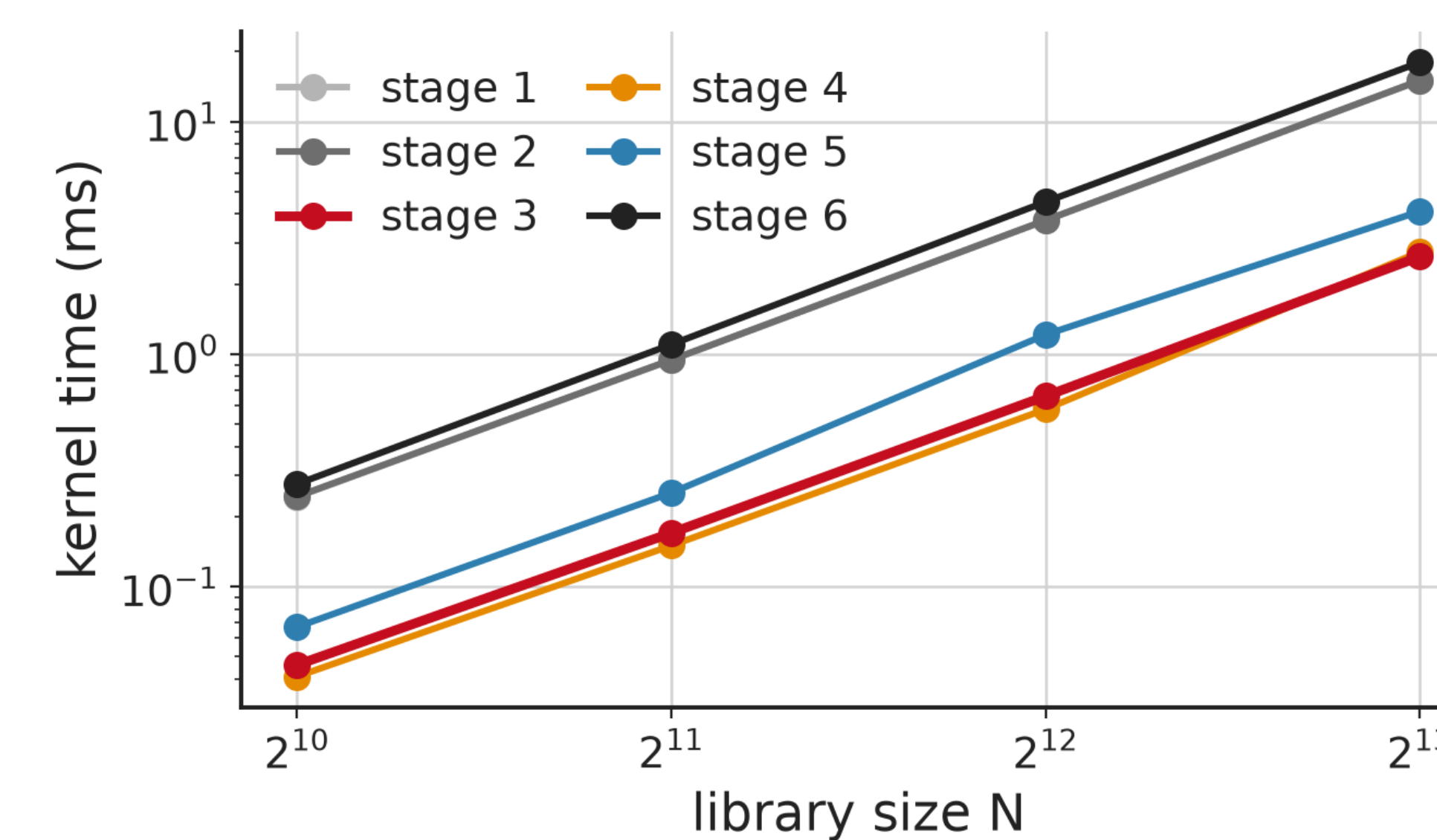
RESULTS

Measured on NVIDIA RTX 3080 Ti (12 GB), CUDA 13.

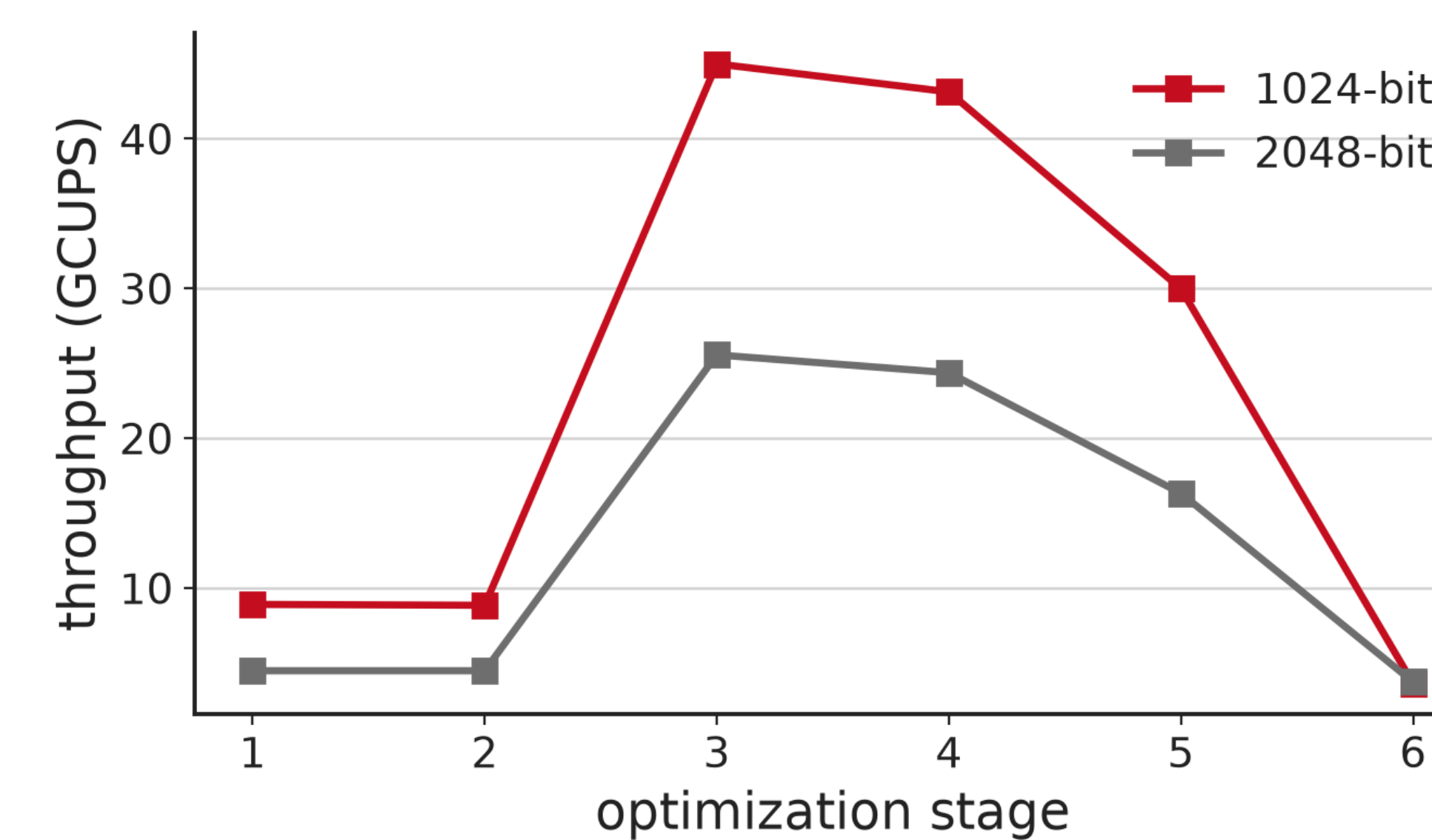
~45 GCUPS

peak throughput --- 1024-bit, coalesced kernel

Runtime grows as $O(N^2)$; the optimized kernel stays well below the naive one.



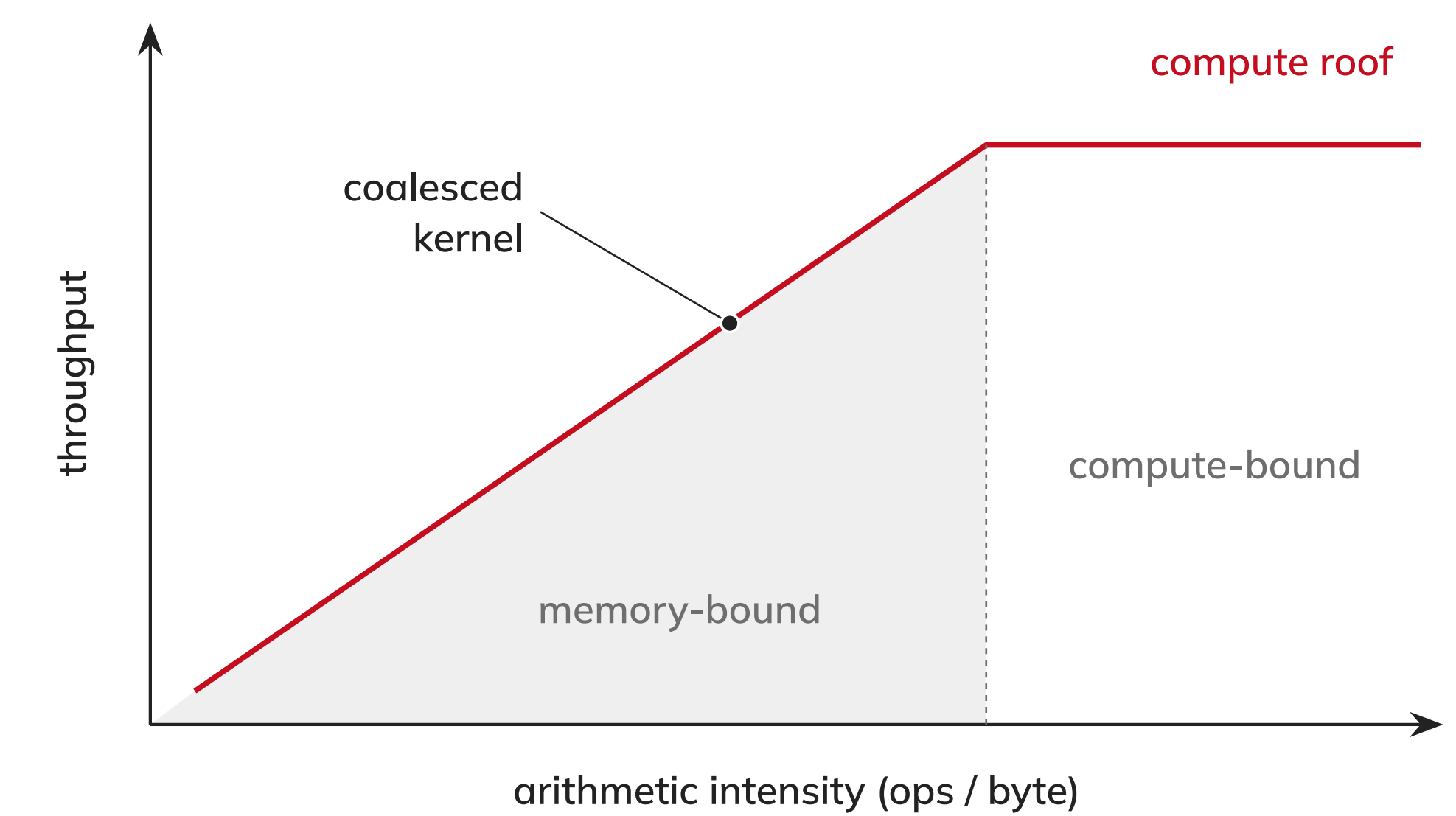
Throughput peaks at coalescing, then drops as the later stages regress.



CONCLUSION

A correct, staged CUDA engine. Coalescing gave the main gain; the kernel then hits a memory-bound ceiling that later stages cannot pass at this size.

5.7x over the naive baseline



The kernel sits on the memory roof: throughput tracks bandwidth, not compute. Coalescing uses bandwidth better and wins; on-chip reuse cannot lift a memory-bound kernel at this size.

REFERENCES

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CODE & DATA

github.com/Kiarashmo/chemsim-gpu

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